SARATHKUMAR DUNNAPAPAGARI

Aspiring Cloud & ML Engineer · M.Tech CSE Student · B.Tech IT Graduate

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PROFILE SUMMARY

Motivated M.Tech Computer Science student and B.Tech IT graduate with four internships and trainings in AWS cloud, Python development and data science / machine learning. Hands-on experience building real-time computer vision, machine learning and full-stack projects with Python, PyTorch, YOLOv8, FastAPI and React, and deploying on AWS and GCP. Eager to start my career in a full-time role where I can learn fast and build reliable, scalable solutions.

TECHNICAL SKILLS

Languages	Python, Java, C
Cloud	AWS EC2, S3, IAM, Google Cloud (GCP)
ML & Data	PyTorch, scikit-learn, YOLOv8, OpenCV, Pandas, DuckDB
Web & APIs	FastAPI, React, HTML, CSS, JavaScript
Tools	Docker, Git & GitHub, Prometheus, Grafana, Linux, MS Office, AutoCAD

PROJECTS

Construction Site Safety Monitor GitHub · Project page — **Computer Vision · Cloud**

- Built a real-time YOLOv8 pipeline for webcam, CCTV or video files that streams live detections to a React dashboard.
- Designed PPE-ready alerts: helmet / vest violations with S3 snapshots switch on once a custom PPE model is configured.
- Containerized with Docker Compose and NGINX, with an EC2 setup script and Prometheus + Grafana monitoring.

Stack: Python · YOLOv8 · OpenCV · FastAPI · React + TypeScript · Docker · AWS S3 · Prometheus · Grafana

Data Analyst + ML Agent GitHub — **Full-stack · ML**

- Developed an AI-assisted data analysis platform: explore datasets with charts, plain-English questions or SQL, then train and compare 11 ML algorithms (5 classification, 6 regression).
- Added explainability, PDF reports, optional LLM explanations and 14 automated backend tests; works fully without an API key.

Stack: FastAPI · Pandas · DuckDB · scikit-learn · React · TypeScript · Tailwind CSS

EmoMusic GitHub — **Deep Learning**

- Created a music recommender driven by real-time facial emotion recognition that eases the listener toward a calmer mood.
- Fine-tuned an EfficientNet-B0 classifier on FER-2013 for 7 emotions, benchmarked it against a baseline CNN, and smoothed predictions to keep the playlist stable.

Stack: PyTorch · timm · OpenCV

EDUCATION

M.Tech, Computer Science and Engineering GMR Institute of Technology, Rajam *Pursuing* — **2025 – 2027**

B.Tech, Information Technology Sir C.R. Reddy College of Engineering, Eluru **CGPA 7.29** — **2021 – 2024**

Diploma, Civil Engineering M.R.A.G.R Government Polytechnic, Vizianagaram **67.69%** — **2017 – 2020**

INTERNSHIPS

AWS Cloud Intern BrainOVision Solutions — **2 months**

- Deployed apps on AWS EC2, managed S3 storage, explored IAM roles, and set up and monitored cloud infrastructure.

Data Science & Machine Learning Intern APSCHE — **6 months**

- Worked on predictive modeling, applying ML algorithms and improving accuracy through data preprocessing.

Python Developer Intern BrainOVision Solutions — **2 months**

- Developed Python automation scripts and small applications, improving efficiency with optimized code.

Industrial Training APSPHC — **6 months**

- Gained exposure to infrastructure and technical workflows in public sector projects.

CERTIFICATIONS

- **Google Cloud Computing Foundations** — NPTEL
- **Getting Started with Amazon Web Services** — SkillUp
- **Cloud Computing** — NPTEL
- **FDP on Cloud Infrastructure** — BrainOVision
- **Introduction to Soft Skills** — TCS